Executive Summary

- **Use case:** Build an end-to-end Snowflake data engineering solution on Electric Vehicle population data that supports analytics, data sharing, and natural-language business access.
- **Approach:** Snowflake-native medallion architecture with raw JSON landing in Bronze, validated and deduplicated conformed data in Silver, and business-ready marts in Gold.
- **Differentiator:** Add semantic analytics with Cortex Analyst + Streamlit so business users can ask natural-language questions without writing SQL.
- **Open architecture:** Use Apache Iceberg selectively where interoperability matters, while keeping core curated analytics optimized for Snowflake performance and governance.

Use Case Description

Business problem The organization needs a production-style data engineering pattern that turns semi-structured EV population data into trusted, shareable, analytics-ready assets for both technical and business users.

Audience

- Data engineers need governed ingestion, transformation, quality controls, and orchestration.
- Architects need a scalable pattern with interoperability and data sharing.
- ML engineers need curated features and reusable dimensions/facts.
- Directors and VPs need fast answers on EV growth, regional adoption, incentives, and manufacturer trends..

Questions the platform should answer

- Where is EV adoption growing fastest?
- Which makes and models dominate by county or utility territory?
- What share of vehicles qualify for CAFV incentives?
- How is BEV vs PHEV mix changing over time?

Dataset and Analytical Scope

- Source shape: The demo exercise uses the Electric Vehicle Population JSON dataset and asks for field extraction, insights, medallion layering, quality checks, orchestration, sharing, and semantic access.

Numeric Column — MODEL_YEAR:

Stat	Value
count	22,183
mean	2019
std	2.95
min	1999
25%	2017
50%	2020
75%	2022
max	2023

Categorical Columns:

Column	count	unique	freq
SOURCE_DATASET_ID	22,183	1	22,183
DATA_PROVIDER	22,183	1	22,183
VIN	22,183	4,986	91
COUNTY	22,182	79	11,618
CITY	22,182	386	3,993
STATE	22,183	19	22,120
MAKE	22,183	33	9,917
MODEL	22,151	110	4,252
EV_TYPE	22,183	2	17,060
LOCATION_POINT	22,176	480	609

Validation summary:

Check	Result
Row count match	22,183 (source) = 22,183 (iceberg)
source_dataset_id	22,183 non-null
vin	22,183 non-null
make	22,183 non-null
model	22,151 non-null (32 nulls — likely missing data in source)
model_year range	1999–2023
ingested_at	22,183 non-null
Distinct states	19

Here's the null analysis:

Column	Nulls	Assessment
county	1	Single TX row missing county/city/location — likely incomplete registration data. Could look up by VIN or leave as-is.
city	1	Same row as above (VIN `7SAYGDEE3N`).
location_point	7	6 have county/city/state but no coordinates; 1 (TX) is missing everything geographic except state.
All other columns	0	Clean.

Imputation strategy recommendations:

1. **county & city** (1 null each) — Same record. Could impute via reverse-lookup from other rows with the same VIN prefix or state, but with only 1 row it's safer to leave as `'UNKNOWN'` or investigate the source.
2. **location_point** (7 nulls) — These are geographic coordinates. Options:
   - **Geocode from city/state**: For the 6 rows that have city+county, derive approximate coordinates from a geocoding reference or from the mode of `location_point` for matching city+state combinations.
   - **Leave null**: The TX row has no city/county, so geocoding isn't feasible without external data.
   - **Flag as missing**: Add a `location_imputed` boolean column if you geocode.

Validation Results for `SILVER_EV_CLEANED`:

Check	Result
Total rows	22,183
Null model count	0
Volvo 2023 BEV - C40	34 (was 2 original + 32 imputed)
Volvo 2023 BEV - XC40	1 (unchanged)

All 32 null `MODEL` values were successfully imputed with **C40** (the mode for Volvo 2023 BEV). Zero nulls remain.

Recommended Architecture Overview

Design principles

- Keep raw immutable data for auditability.
- Separate cleansing from business semantics.
- Materialize Gold for high-value repeated consumption.
- Use Snowflake-native features first; introduce open table formats only where they solve a real interoperability need.

Core services

- Snowflake tables / stages / file formats for ingestion
- SQL/dbt for transformations
- Dynamic Tables for automated refresh
- Tasks / orchestration for dependency execution
- Iceberg tables for open table interoperability
- Secure Data Sharing for internal and cross-org consumption
- Cortex Analyst + semantic view + Streamlit for natural-language analytics

Pattern:
JSON files → Bronze (raw) → [Dynamic Table] → Silver (cleaned) → [Dynamic Table] → Gold (aggregated) Semantic model → Streamlit / sharing / downstream consumers

Medallion Architecture Deep Dive

Bronze

- Store raw JSON exactly as received.
- Preserve metadata such as file name, load timestamp, source, and ingestion batch.
- Parse JSON into typed columns.
- **Purpose:** replayability, audit trail, schema drift tolerance.

Silver

- Standardize values, handle nulls, validate schema, deduplicate, and enforce business rules.
- Add technical keys, conformance logic, and lineage attributes.
- **Purpose:** trusted reusable foundation for multiple use cases.

Gold

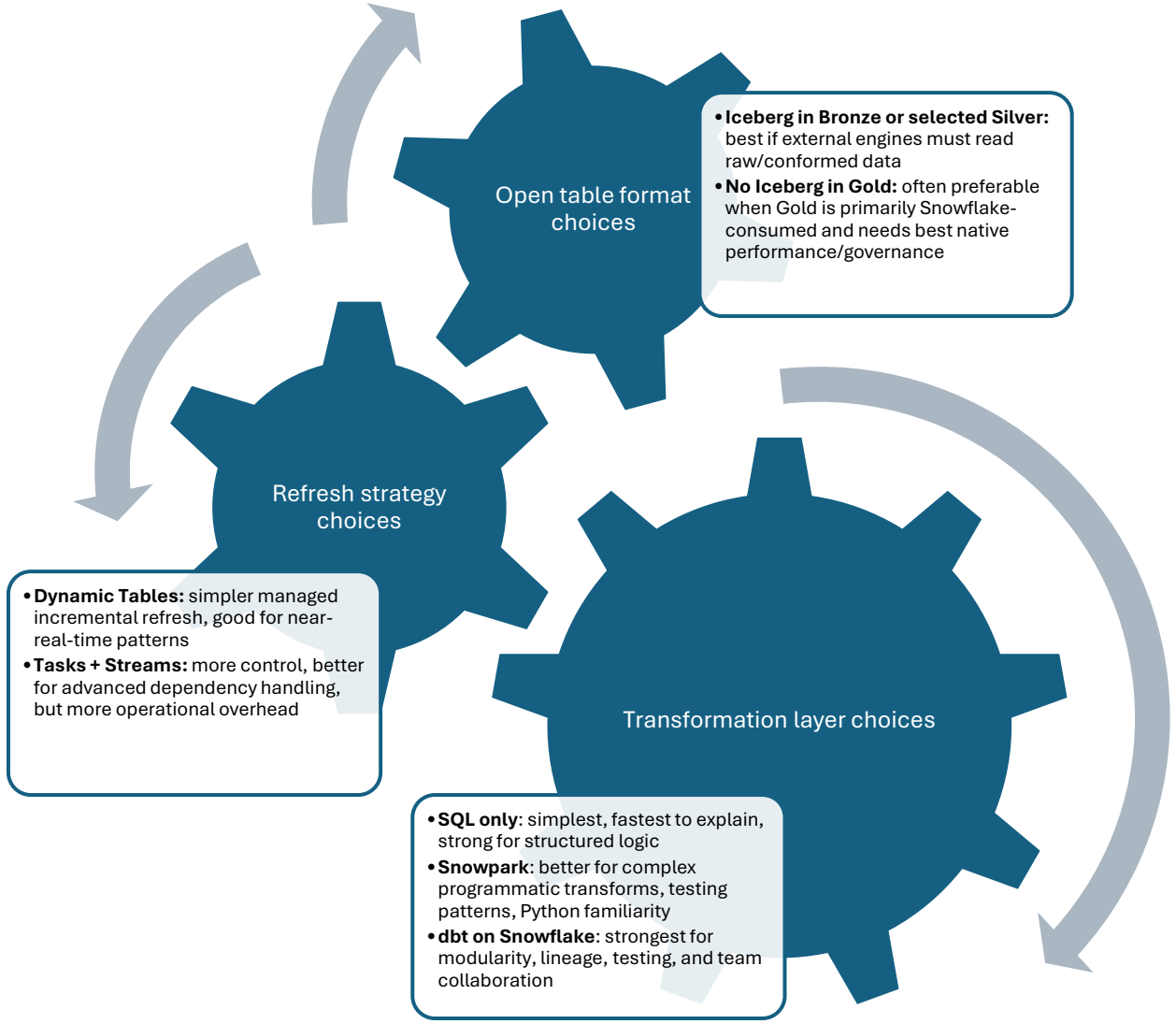
- Build dimensions and facts for business consumption.
- Create aggregates by geography, make/model, EV type, CAFV eligibility, and time.
- Expose semantic-ready views and shareable curated datasets.
- **Purpose**: fast access, simple business language, stable contracts for downstream users.

All checks pass — perfect match across every metric.

Validation Check	Silver	Gold	Match
Total vehicles	22,183	22,183	✓
Distinct states	19	19	✓
Distinct model years	17	17	✓
BEV count	17,060	17,060	✓
PHEV count	5,123	5,123	✓
Per-state diff	—	—	All 19 states = 0 diff

No data loss or duplication in the gold layer aggregation.

Trade-offs and Alternatives



Business Value vs Traditional Spark/Lakehouse

Where does the proposed solution create advantage?

- Fewer moving parts: storage, compute, governance, sharing, app layer, and semantic analytics in one platform
- Faster time to usable insights for mixed technical and business audiences
- Lower operational burden than managing Spark clusters, notebooks, meta-stores, and multiple security surfaces
- Native data sharing reduces extract-copy-distribute patterns
- Cortex Analyst shortens distance from curated data to business consumption

For this EV analytics use case, this solution is stronger when speed, governance, business accessibility, and reduced complexity matter more than maximizing ecosystem optionality.

Business Outcomes

For business leaders

- Faster visibility into EV market trends and incentive opportunities
- More self-service analytics with less dependence on analysts for every question
- Better trust through governed, curated, explainable data products

For the organization

- Cleaner medallion separation and better reusability
- Lower ops complexity than a fragmented lake-house stack
- Easier path to productionizing semantic analytics and data sharing

Build the EV solution as a Snowflake-native medallion architecture with selective Iceberg interoperability, strong data quality controls, dynamic refresh, governed sharing, and Cortex Analyst-driven semantic access.